

The Uber Assumption and its Random Self-Reducibility

in Non-Bilinear and Type 1, 2, 3 Bilinear Groups

Abstract

We fix notation for prime-order groups without a pairing and for bilinear groups in each of the three Galbraith–Paterson–Smart types, formally define the Uber assumption in all four settings, and treat random self-reducibility in each. A single generic-computability lemma isolates, per setting, the space of exponents a reduction can realize; a single affine-reparametrization theorem then yields random self-reducibility whenever the source data are stable under the reparametrization and the challenge polynomial transforms covariantly. The four settings differ only through the product structure the pairing (and, in Type 2, the homomorphism ψ) makes available, and this difference is exactly what enlarges or shrinks the self-reducible subclass. We give a worked self-reducible assumption in each setting and exhibit a target that is self-reducible in Type 2 but not, by any affine reparametrization, in Type 3. A final section places the theorem against classical random self-reducibility: the affine condition is the one-query case of the non-adaptive condition of Feigenbaum and Fortnow, and the classical multi-query interpolation of Lipton, which the group encoding admits unchanged, does reach the separating target — but only its computational form, and only against a high-advantage solver. The two techniques are incomparable, and the decisional, negligible-advantage regime that cryptography works in is the one where only the affine route survives.

1 Introduction

The discrete-logarithm problem is random self-reducible by the shift $g^x \mapsto g^x g^r$ [TW87]; the answer for a worst-case instance is recovered from an average-case solver by subtracting r . This note carries the underlying idea, inject a uniform reparametrization of the secret and undo it on the answer, across the standard group settings and states it uniformly. Throughout, p is a prime and all groups have order p . We use implicit (bracket) notation in the style of [EHKRV13]. Section 10 relates the resulting condition to the classical form of the same idea, in which the undoing is done by polynomial interpolation across several correlated queries rather than in closed form on one [Lipton91, FF93].

2 Group settings

2.1 Non-bilinear groups

Definition 1 (prime-order group). A *(non-bilinear) group generator* outputs $\mathcal{G} = (p, \mathbb{G}, g)$ where $\mathbb{G}$ is a cyclic group of prime order p , g a generator, and the group law and equality are computable in probabilistic polynomial time. For $a \in \mathbb{Z}_p$ write $[a] := g^a$, so $[a] \cdot [b] = [a + b]$ and $[a]^c = [ca]$. There is no pairing.

2.2 Bilinear groups and the GPS taxonomy

Definition 2 (bilinear group). A *bilinear group generator* outputs $\mathcal{BG} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e, g_1, g_2)$ where $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$ are cyclic of prime order p , g_1, g_2 generate $\mathbb{G}_1, \mathbb{G}_2$, all group laws and equalities are efficient, and $e: \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is an efficiently computable, non-degenerate bilinear map. For $a \in \mathbb{Z}_p$ write $[a]_1 := g_1^a$, $[a]_2 := g_2^a$, and $[a]_T := e(g_1, g_2)^a$, so that

$$e([a]_1, [b]_2) = [ab]_T, \quad [a]_i \cdot [b]_i = [a + b]_i, \quad [a]_i^c = [ca]_i.$$

Following Galbraith, Paterson, and Smart [GPS08], bilinear groups come in three types, distinguished by the efficiently computable isomorphisms between $\mathbb{G}_1$ and $\mathbb{G}_2$:

Type 1. $\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}$ and e is symmetric, $e(u, v) = e(v, u)$. One source group.

Type 2. $\mathbb{G}_1 \neq \mathbb{G}_2$, with an efficiently computable isomorphism $\psi: \mathbb{G}_2 \rightarrow \mathbb{G}_1$ (so $\psi([a]_2) = [a]_1$), but none known in the reverse direction.

Type 3. $\mathbb{G}_1 \neq \mathbb{G}_2$, with no efficiently computable isomorphism in either direction.

Type 3 is the most efficient and most used in practice; Type 1 is the most permissive algebraically. The only structural difference for what follows is which group elements can be moved between $\mathbb{G}_1$ and $\mathbb{G}_2$, and hence which products the pairing can form.

3 The generic-computability calculus

Fix a setting and an instance consisting of encodings of finite polynomial tuples $R \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_1$, $S \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_2$, and $T \subseteq \mathbb{Z}_p[\vec{X}]$ in $\mathbb{G}_T$ (in a single-group setting only R and, if applicable, T occur). Each tuple is assumed to contain the constant 1, since the generators are public. For $\vec{X} = (X_1, \dots, X_n)$ and a secret $\vec{x} \in \mathbb{Z}_p^n$, the instance is the collection of $[r_i(\vec{x})]_1, [s_j(\vec{x})]_2, [t_k(\vec{x})]_T$.

For a group $\gamma \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$ (or $\mathbb{G}$) let $\mathcal{L}_\gamma \subseteq \mathbb{Z}_p[\vec{X}]$ denote the polynomials q for which $[q(\vec{x})]_\gamma$ is computable in probabilistic polynomial time from the instance and public parameters, uniformly in $\vec{x}$. Write $A \cdot B = \{ab : a \in A, b \in B\}$ and let $\langle \cdot \rangle$ be $\mathbb{Z}_p$ -linear span.

Lemma 1 (computable exponents). *With the conventions above,*

$$\text{non-bilinear: } \mathcal{L}_{\mathbb{G}} = \langle R \rangle;$$

$$\text{Type 1 } (\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}, \text{ source tuple } R) : \mathcal{L}_{\mathbb{G}} = \langle R \rangle, \quad \mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T \rangle;$$

$$\begin{aligned} \text{Type 2 } (\psi : \mathbb{G}_2 \rightarrow \mathbb{G}_1) : \quad \mathcal{L}_1 = \langle R \cup S \rangle, \quad \mathcal{L}_2 = \langle S \rangle, \\ \mathcal{L}_T = \langle (R \cup S) \cdot S \cup T \rangle; \end{aligned}$$

$$\text{Type 3: } \mathcal{L}_1 = \langle R \rangle, \quad \mathcal{L}_2 = \langle S \rangle, \quad \mathcal{L}_T = \langle R \cdot S \cup T \rangle.$$

The inclusions “ $\supseteq$ ” hold unconditionally: every listed polynomial’s encoding is computable by group operations and pairings. The reverse inclusions hold in the generic bilinear group model, where these are exactly the computable encodings [BBGo5, Boyeno8].

Justification. The group law realizes all $\mathbb{Z}_p$ -linear combinations, giving $\langle \cdot \rangle$ in each group. The pairing sends a computable $\mathbb{G}_1$ -exponent $a \in \mathcal{L}_1$ and a computable $\mathbb{G}_2$ -exponent $b \in \mathcal{L}_2$ to $ab \in \mathcal{L}_T$; pairing against the generator ($b = 1$ or $a = 1$) shows $\mathcal{L}_1, \mathcal{L}_2 \subseteq \mathcal{L}_T$. In Type 1 both pairing arguments range over $\langle R \rangle$, producing all products $R \cdot R$. In Type 2, ψ moves each $\mathbb{G}_2$ encoding into $\mathbb{G}_1$, so $\mathcal{L}_1$ gains S , and the pairable products become $\mathcal{L}_1 \cdot \mathcal{L}_2 = (R \cup S) \cdot S$; the missing products $R \cdot R$ would require an element of $\mathbb{G}_2$ with an R -exponent, which ψ cannot supply. In Type 3 neither group maps to the other, so $\mathcal{L}_1 = \langle R \rangle$, $\mathcal{L}_2 = \langle S \rangle$, and only cross products $R \cdot S$ arise. The generic-model converse is the master lemma of [BBGo5, Boyeno8]. $\square$

The single distinction across the four rows is the set of $\mathbb{G}_T$ -products: none (no pairing), $R \cdot R$ (Type 1), $R \cdot S \cup S \cdot S$ (Type 2), $R \cdot S$ (Type 3). Everything below is a consequence of this.

4 The Uber assumption

Definition 3 (bilinear Uber problem). Fix a bilinear setting, source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$), each containing 1, a challenge polynomial $f \in \mathbb{Z}_p[\tilde{X}]$, and a challenge group $\text{ch} \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$; write $\mathcal{L}_\star := \mathcal{L}_{\text{ch}}$ (Lemma 1). For a secret $\vec{x} \in \mathbb{Z}_p^n$ the instance is

$$\text{Inst}(\vec{x}) = (\{[r_i(\vec{x})]_1\}_i, \{[s_j(\vec{x})]_2\}_j, \{[t_k(\vec{x})]_T\}_k).$$

- *Computational* $(R, S, T, f; \text{ch})$ -Uber: given $\text{Inst}(\vec{x})$, output $[f(\vec{x})]_\star$.
- *Decisional* $(R, S, T, f; \text{ch})$ -Uber: given $\text{Inst}(\vec{x})$ and Z , decide whether $Z = [f(\vec{x})]_\star$ (real) or $Z \leftarrow_{\S} \text{ch}$ (random).

The *average case* draws $\vec{x} \leftarrow_{\S} \mathbb{Z}_p^n$; the *worst case* quantifies over all $\vec{x}$. For a decision algorithm $\mathcal{A}$ put

$$\delta(\mathcal{A}; \vec{x}) = \Pr[\mathcal{A}(\text{Inst}(\vec{x}), [f(\vec{x})]_\star) = 1] - \Pr[\mathcal{A}(\text{Inst}(\vec{x}), U) = 1], \quad U \leftarrow_{\S} \text{ch},$$

and $\delta_{\text{avg}}(\mathcal{A}) = \mathbb{E}_{\vec{x} \leftarrow_{\S} \mathbb{Z}_p^n} [\delta(\mathcal{A}; \vec{x})]$; for a search algorithm put $\text{Succ}(\mathcal{A}; \vec{x}) = \Pr[\mathcal{A}(\text{Inst}(\vec{x})) = [f(\vec{x})]_\star]$ and $\text{Succ}_{\text{avg}}(\mathcal{A}) = \mathbb{E}_{\vec{x}} [\text{Succ}(\mathcal{A}; \vec{x})]$. The assumption

asserts that δ_{avg} (resp. Succ_{avg}) is negligible for all probabilistic polynomial-time $\mathcal{A}$.

The four settings are Definition 3 with the computable spaces of Lemma 1. We record each explicitly.

Definition 4 (non-bilinear Uber). Setting $\mathcal{G} = (p, \mathbb{G}, g)$; source tuple R (in $\mathbb{G}$, containing 1); challenge $f \in \mathbb{Z}_p[\tilde{X}]$ in $\mathbb{G}$. Instance $\text{Inst}(\vec{x}) = \{[r_i(\vec{x})]\}_i$; goal, compute or distinguish $[f(\vec{x})]$. Computable space $\mathcal{L}_\star = \mathcal{L}_{\mathbb{G}} = \langle R \rangle$; no pairing, hence no products.

Definition 5 (Type-1 Uber). Setting $\mathcal{BG}$ of Type 1, $\mathbb{G}_1 = \mathbb{G}_2 = \mathbb{G}$; source tuple R (in $\mathbb{G}$), target tuple T (in $\mathbb{G}_T$), each containing 1; challenge f in $\text{ch} \in \{\mathbb{G}, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_{\mathbb{G}} = \langle R \rangle$ and $\mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T \rangle$.

Definition 6 (Type-2 Uber). Setting $\mathcal{BG}$ of Type 2 with $\psi: \mathbb{G}_2 \rightarrow \mathbb{G}_1$; source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$); challenge f in $\text{ch} \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_1 = \langle R \cup S \rangle$, $\mathcal{L}_2 = \langle S \rangle$, $\mathcal{L}_T = \langle (R \cup S) \cdot S \cup T \rangle$.

Definition 7 (Type-3 Uber). Setting $\mathcal{BG}$ of Type 3; source tuples R (in $\mathbb{G}_1$), S (in $\mathbb{G}_2$), T (in $\mathbb{G}_T$); challenge f in $\text{ch} \in \{\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T\}$. Computable spaces $\mathcal{L}_1 = \langle R \rangle$, $\mathcal{L}_2 = \langle S \rangle$, $\mathcal{L}_T = \langle R \cdot S \cup T \rangle$.

Remark 1 (hardness). The master theorem of [BBGo5, Boyeno8] states, informally, that the decisional $(R, S, T, f; \text{ch})$ -Uber problem is hard in the generic bilinear group model if and only if $f \notin \mathcal{L}_\star$, that is, the challenge exponent is linearly independent of the computable ones. Random self-reducibility is a separate property and does not require hardness; the two interact through Lemma 1, as discussed in Section 11.

5 An affine random-self-reducibility theorem

Let $\text{AGL}_n(\mathbb{Z}_p)$ be the affine group of maps $\tau_{A, \vec{b}}: \vec{x} \mapsto A\vec{x} + \vec{b}$ with $A \in \text{GL}_n(\mathbb{Z}_p)$, acting on polynomials by $(q \circ \tau)(\vec{X}) = q(A\vec{X} + \vec{b})$. The following theorem is setting-agnostic: it refers only to the computable spaces of Lemma 1.

Theorem 1 (master reduction). *Consider any of the four settings, an instance with source tuples R, S, T and challenge $(f; \text{ch})$, and computable spaces $\{\mathcal{L}_\gamma\}$ with $\mathcal{L}_\star = \mathcal{L}_{\text{ch}}$. Suppose there is an efficiently sampleable distribution $\mathcal{D}$ on a subgroup $\mathcal{T} \leq \text{AGL}_n(\mathbb{Z}_p)$ such that, for every $\tau \in \mathcal{T}$,*

- (C1) $r_i \circ \tau \in \mathcal{L}_1$, $s_j \circ \tau \in \mathcal{L}_2$, and $t_k \circ \tau \in \mathcal{L}_T$ for all i, j, k (with $\mathcal{L}_1 = \mathcal{L}_{\mathbb{G}}$ in the single-group settings);
- (C2) $f \circ \tau = \lambda_\tau f + h_\tau$ for a scalar $\lambda_\tau \in \mathbb{Z}_p^\times$ and a polynomial $h_\tau \in \mathcal{L}_\star$, both computable from τ and the parameters;

and, moreover,

(C3) $\tau(\vec{x})$ is uniform on $\mathbb{Z}_p^n$ for every fixed $\vec{x}$ when $\tau \leftarrow \mathcal{D}$.

Then the computational and decisional $(R, S, T, f; \text{ch})$ -Uber problems are random self-reducible: there is a probabilistic polynomial-time reduction $\mathcal{B}^{\mathcal{A}}$, using one draw from $\mathcal{D}$ and one oracle call, with $\text{Succ}(\mathcal{B}; \vec{x}_0) = \text{Succ}_{\text{avg}}(\mathcal{A})$ and $\delta(\mathcal{B}; \vec{x}_0) = \delta_{\text{avg}}(\mathcal{A})$ for every $\vec{x}_0 \in \mathbb{Z}_p^n$.

Proof. On input $\text{Inst}(\vec{x}_0)$ (and a challenge Z in the decisional case) the reduction proceeds as follows.

Reparametrize. Draw $\tau \leftarrow \mathcal{D}$ and set $\vec{x}' := \tau(\vec{x}_0)$, which is unknown to $\mathcal{B}$ but uniform on $\mathbb{Z}_p^n$ by (C3).

Rebuild the instance. By (C1), each rebuilt exponent lies in the computable space of its group, so by the “ $\supseteq$ ” part of Lemma 1 its encoding is computable from $\text{Inst}(\vec{x}_0)$. Concretely, writing $r_i \circ \tau = \sum_a c_a g_a$ as a $\mathbb{Z}_p$ -combination of the generators of $\mathcal{L}_1$, one has $[r_i(\vec{x}')]_1 = [(r_i \circ \tau)(\vec{x}_0)]_1 = [\prod_a [g_a(\vec{x}_0)]_1^{c_a}]_1$, and similarly for S (using ψ in Type 2 to realize any S -exponent in $\mathbb{G}_1$) and for T (using pairings $e([\cdot]_1, [\cdot]_2)$ to realize products). This yields $\text{Inst}(\vec{x}')$, distributed exactly as a fresh average-case instance.

Transport the challenge. By (C2) compute λ_τ, h_τ and the element $[h_\tau(\vec{x}_0)]_\star$ (computable since $h_\tau \in \mathcal{L}_\star$).

Decisional case. Set $Z' := Z^{\lambda_\tau} \cdot [h_\tau(\vec{x}_0)]_\star$ and output $\mathcal{A}(\text{Inst}(\vec{x}'), Z')$. If $Z = [f(\vec{x}_0)]_\star$ then

$$Z' = [\lambda_\tau f(\vec{x}_0) + h_\tau(\vec{x}_0)]_\star = [(f \circ \tau)(\vec{x}_0)]_\star = [f(\vec{x}')]_\star,$$

the real challenge for $\vec{x}'$. If Z is uniform on ch , then Z^{λ_τ} is uniform because $z \mapsto z^{\lambda_\tau}$ is a bijection of the order- p cyclic group ch (as $\gcd(\lambda_\tau, p) = 1$), so Z' is uniform and independent of $\text{Inst}(\vec{x}')$. Writing p_1, p_0 for $\mathcal{A}$'s acceptance probabilities on real and random challenges,

$$\delta(\mathcal{B}; \vec{x}_0) = \mathbb{E}_\tau[p_1(\tau(\vec{x}_0))] - \mathbb{E}_\tau[p_0(\tau(\vec{x}_0))] \stackrel{(C3)}{=} \mathbb{E}_{\vec{x}'}[p_1] - \mathbb{E}_{\vec{x}'}[p_0] = \delta_{\text{avg}}(\mathcal{A}).$$

Search case. Run $W \leftarrow \mathcal{A}(\text{Inst}(\vec{x}'))$. On success $W = [f(\vec{x}')]_\star = [\lambda_\tau f(\vec{x}_0) + h_\tau(\vec{x}_0)]_\star$, so output $Y := (W \cdot [h_\tau(\vec{x}_0)]_\star^{-1})^{\lambda_\tau^{-1} \bmod p} = [f(\vec{x}_0)]_\star$. Since $\vec{x}'$ is uniform, the success probability equals $\text{Succ}_{\text{avg}}(\mathcal{A})$. $\square$

Remark 2 (relaxing (C3)). When $\mathcal{D}$ makes $\tau(\vec{x})$ uniform only on a subset $\Omega \subseteq \mathbb{Z}_p^n$ (for instance $\mathbb{Z}_p^\times \times \mathbb{Z}_p$ under a multiplicative reparametrization of one coordinate), the transfer holds with the average taken over Ω . If the complement $\mathbb{Z}_p^n \setminus \Omega$ is a proper affine subvariety on which $f(\vec{x})$ takes a value the reduction can compute directly from $\text{Inst}(\vec{x}_0)$ (a common case is $f(\vec{x}) = 0$ there), the reduction detects that case by an equality test on the given encodings and answers it without an oracle call, recovering full worst-case coverage.

6 Random self-reducibility: non-bilinear groups

Here $\mathcal{L}_\star = \mathcal{L}_{\mathbb{G}} = \langle R \rangle$ and no pairing exists, so (C2) demands the covariance defect to be a $\mathbb{Z}_p$ -combination of the source exponents themselves. The motivating scalar

case is discrete log: to solve g^x query the solver on g^{x+r} and subtract r [TW87]; its group-valued cousins fit Definition 3.

Proposition 1 (CDH and DDH). *Let $R = (1, X_1, X_2)$, so $\mathcal{L}_{\mathbb{G}} = \langle 1, X_1, X_2 \rangle$, and $f = X_1 X_2$ with challenge in $\mathbb{G}$. The translation group $\mathcal{T} = \{\vec{x} \mapsto \vec{x} + \vec{b}\}$ with $\vec{b} \leftarrow_{\S} \mathbb{Z}_p^2$ satisfies (C1)–(C3), so computational CDH and decisional DDH are random self-reducible.*

Proof. $X_i \circ \tau = X_i + b_i \in \mathcal{L}_{\mathbb{G}}$ gives (C1). For (C2), $f \circ \tau = (X_1 + b_1)(X_2 + b_2) = f + (b_2 X_1 + b_1 X_2 + b_1 b_2)$, so $\lambda_{\tau} = 1$ and $h_{\tau} = b_2 X_1 + b_1 X_2 + b_1 b_2 \in \langle 1, X_1, X_2 \rangle = \mathcal{L}_{\star}$. Translation gives exact uniformity, (C3). Explicitly the reduction returns $[f(\vec{x}_0)] = W \cdot [x_1]^{-b_2} [x_2]^{-b_1} [1]^{-b_1 b_2}$ in the search case. $\square$

With $R = (1, X_1, \dots, X_n)$ we have $\mathcal{L}_{\star} = V_{\leq 1}$ (affine polynomials), and translating a degree- d monomial leaves monomials of degree at most $d - 1$; hence (C2) holds under translation exactly when $\deg f \leq 2$. Without a pairing, degree 2 is the ceiling: a product target is the most that is randomizable, and it is.

7 Random self-reducibility: Type 1

Here $\mathcal{L}_{\star} = \mathcal{L}_{\mathbb{G}_T} = \langle R \cdot R \cup T \rangle$: the pairing supplies all products of source exponents, so the covariance defect may use any degree-2 combination of them.

Proposition 2 (BDDH). *Let $R = (1, X_1, X_2, X_3)$, $T = (1)$, so $\mathcal{L}_{\mathbb{G}_T} = V_{\leq 2}$, and $f = X_1 X_2 X_3$ with challenge in $\mathbb{G}_T$. Translation satisfies (C1)–(C3), so decisional and computational BDDH are random self-reducible.*

Proof. $X_i \circ \tau = X_i + b_i \in \mathcal{L}_{\mathbb{G}} = \langle R \rangle$ gives (C1). Expanding,

$$f \circ \tau = X_1 X_2 X_3 + [b_3 X_1 X_2 + b_2 X_1 X_3 + b_1 X_2 X_3 + (\deg \leq 1)],$$

so $\lambda_{\tau} = 1$ and h_{τ} has total degree at most 2, hence lies in $V_{\leq 2} = \mathcal{L}_{\star}$; each of its monomials is realized by a single pairing, for instance $[x_1 x_2]_T = e([x_1]_1, [x_2]_2)$. Translation gives (C3). $\square$

With coordinate source $R = (1, X_1, \dots, X_n)$ we get $\mathcal{L}_{\star} = V_{\leq 2}$, so the same degree count gives translation-randomizability for every f with $\deg f \leq 3$. The pairing has raised the ceiling from 2 (no pairing) to 3.

8 Random self-reducibility: Type 2

Here $\mathcal{L}_1 = \langle R \cup S \rangle$ and $\mathcal{L}_T = \langle (R \cup S) \cdot S \cup T \rangle$: relative to Type 3 the homomorphism ψ adds the right-hand products $S \cdot S$ to the target space. This extra room is what makes the following target randomizable.

Proposition 3. *In a Type-2 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and $f = X_1 X_2^2$ with challenge in $\mathbb{G}_T$. Then $f \notin \mathcal{L}_{\star}$ (so the assumption is generically hard), and translation satisfies (C1)–(C3); the problem is random self-reducible.*

Proof. Here $\mathcal{L}_T = \langle (R \cup S) \cdot S \rangle = \langle 1, X_1, X_2, X_1 X_2, X_2^2 \rangle$, and $X_1 X_2^2$ is not in this span, giving generic hardness. Under $\tau: \vec{x} \mapsto \vec{x} + \vec{b}$ one has $X_1 \circ \tau = X_1 + b_1 \in \mathcal{L}_1 = \langle 1, X_1 \rangle$ and $X_2 \circ \tau = X_2 + b_2 \in \mathcal{L}_2 = \langle 1, X_2 \rangle$, which is (C1). Expanding,

$$f \circ \tau = (X_1 + b_1)(X_2 + b_2)^2 = f + \underbrace{2b_2 X_1 X_2 + b_2^2 X_1 + b_1 X_2^2 + 2b_1 b_2 X_2 + b_1 b_2^2}_{h_\tau},$$

so $\lambda_\tau = 1$ and every monomial of h_τ lies in $\mathcal{L}_\star$. The term X_2^2 is realized using $\psi: [x_2^2]_T = e(\psi([x_2]_2), [x_2]_2)$. Translation gives (C3). $\square$

The point is precisely the X_2^2 term of h_τ : it is a product of two $\mathbb{G}_2$ exponents, which the pairing can form only after ψ has moved one of them into $\mathbb{G}_1$. In Type 1 this term is free; in Type 3 it is unavailable, as the next section shows.

9 Random self-reducibility: Type 3

Here $\mathcal{L}_1 = \langle R \rangle$, $\mathcal{L}_2 = \langle S \rangle$, and $\mathcal{L}_T = \langle R \cdot S \cup T \rangle$: the target space contains only cross products. This is the most restrictive setting, and reparametrizations must respect which coordinates are reachable in the challenge group.

Proposition 4 (co-CDH). *In a Type-3 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, and $f = X_1 X_2$ with challenge in $\mathbb{G}_1$. Then $f \notin \mathcal{L}_\star$ (generically hard), and the reparametrization $\tau: (x_1, x_2) \mapsto (\alpha x_1, x_2 + b_2)$ with $\alpha \leftarrow_{\$} \mathbb{Z}_p^\times$, $b_2 \leftarrow_{\$} \mathbb{Z}_p$ satisfies (C1), (C2) and makes $\tau(\vec{x})$ uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p$; the problem is random self-reducible, with the slice $x_1 = 0$ handled directly.*

Proof. Here $\mathcal{L}_\star = \mathcal{L}_1 = \langle 1, X_1 \rangle$, which contains no $\mathbb{G}_2$ variable; and $X_1 X_2 \notin \langle 1, X_1 \rangle$, giving hardness. For (C1), $X_1 \circ \tau = \alpha X_1 \in \mathcal{L}_1$ and $X_2 \circ \tau = X_2 + b_2 \in \mathcal{L}_2 = \langle 1, X_2 \rangle$. For (C2), $f \circ \tau = \alpha X_1 (X_2 + b_2) = \alpha f + \alpha b_2 X_1$, so $\lambda_\tau = \alpha \in \mathbb{Z}_p^\times$ and $h_\tau = \alpha b_2 X_1 \in \mathcal{L}_\star$. The image $\tau(\vec{x}) = (\alpha x_1, x_2 + b_2)$ is uniform on $\mathbb{Z}_p^\times \times \mathbb{Z}_p$ for every fixed $x_1 \neq 0$. If $x_1 = 0$ then $f(\vec{x}) = 0$ and the answer is the identity of $\mathbb{G}_1$; the reduction detects this by testing $[x_1]_1 = [0]_1$ and outputs the identity, per Remark 2. A multiplicative reparametrization of x_1 is forced because an additive shift $x_1 \mapsto x_1 + b_1$ would put $b_1 X_2$ into h_τ , and $X_2 \notin \mathcal{L}_1$ in Type 3. $\square$

The next result shows the Type-3 restriction genuinely bites: the target of Proposition 3, self-reducible in Type 2, is not affine-randomizable in Type 3.

Proposition 5 (a Type-2 versus Type-3 separation). *In a Type-3 group let $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and $f = X_1 X_2^2$ with challenge in $\mathbb{G}_T$. The assumption is generically hard, and no $\tau \in \text{AGL}_2(\mathbb{Z}_p)$ satisfies (C1)–(C3); hence the affine reduction of Theorem 1 does not apply, in contrast to Type 2 (Proposition 3).*

Proof. Hardness holds as $\mathcal{L}_\star = \mathcal{L}_T = \langle R \cdot S \rangle = \langle 1, X_1, X_2, X_1 X_2 \rangle$ and $X_1 X_2^2 \notin \langle 1, X_1, X_2, X_1 X_2 \rangle$. Suppose $\tau \in \text{AGL}_2(\mathbb{Z}_p)$ satisfies (C1). Condition (C1) for $r = X_1$ forces the first coordinate map $\tau_1 = X_1 \circ \tau$ to lie in $\mathcal{L}_1 = \langle 1, X_1 \rangle$, that is $\tau_1 = e_1 X_1 + e_0$

with no X_2 ; likewise (C1) for $s = X_2$ forces $\tau_2 = aX_2 + a_0$ with no X_1 . Invertibility requires $e_1, a \in \mathbb{Z}_p^\times$. Then

$$f \circ \tau = (e_1X_1 + e_0)(aX_2 + a_0)^2 = (e_1X_1 + e_0)(a^2X_2^2 + 2aa_0X_2 + a_0^2).$$

The coefficient of the monomial X_2^2 (which carries no X_1) is e_0a^2 . Since $X_2^2 \notin \mathcal{L}_\star$ and $\lambda_\tau f = \lambda_\tau X_1 X_2^2$ contributes no pure X_2^2 term, condition (C2) forces $e_0a^2 = 0$, hence $e_0 = 0$. But then $\tau_1 = e_1X_1$, so $\tau(\vec{x})_1 = e_1x_1$, which cannot be uniform on $\mathbb{Z}_p$ for a fixed x_1 (it never realizes uniformity across the value 0), so (C3) fails. Therefore no τ meets (C1)–(C3) simultaneously. $\square$

The obstruction is exactly the one flagged after Proposition 3: the covariance defect necessarily contains X_2^2 , a product of two $\mathbb{G}_2$ exponents. In Type 2 the homomorphism ψ places X_2^2 in $\mathcal{L}_T$ and translation works; in Type 3 it is generically uncomputable, and forcing it out of the defect collapses the reparametrization on the $\mathbb{G}_1$ coordinate. Proposition 5 rules out the single-reparametrization affine technique, which is the tool behind every positive result above, but it does not rule out every reduction. Section 10 shows that a multi-query interpolation reduction, of the kind classical random self-reducibility has used since Lipton [Lipton91], does solve the *computational* problem here; the Type-2 versus Type-3 gap for this target is therefore a gap for the decisional problem and for tight reductions, not a structural one. Ruling out every reduction for the decisional problem would additionally require a lower-bound argument such as a meta-reduction.

10 Relation to classical random self-reducibility

Random self-reducibility long predates its use on group-based assumptions, and the classical technique differs from Theorem 1 in one respect that decides exactly what Proposition 5 does and does not rule out. This section isolates the difference, shows that the two techniques are incomparable, and uses the classical one to reach a target the affine one cannot.

10.1 The multi-query condition

Condition (C2) asks a single reparametrized challenge to be a scalar multiple of the original up to a computable defect. The general *non-adaptive* condition, in the sense of Feigenbaum and Fortnow [FF93], asks instead that the original challenge be a $\mathbb{Z}_p$ -combination of *several* reparametrized ones:

(C2') there are $\tau_1, \dots, \tau_m \in \mathcal{T}$, coefficients $c_1, \dots, c_m \in \mathbb{Z}_p$, and a polynomial $h \in \mathcal{L}_\star$, all computable from the sampled data, with

$$f = \sum_{i=1}^m c_i (f \circ \tau_i) + h.$$

Condition (C2) is the case $m = 1$: from $f \circ \tau = \lambda_\tau f + h_\tau$ one gets $f = \lambda_\tau^{-1}(f \circ \tau) - \lambda_\tau^{-1}h_\tau$, which is (C2') with $c_1 = \lambda_\tau^{-1}$ and $h = -\lambda_\tau^{-1}h_\tau$.

Suppose (C1) and (C3) hold for each τ_i separately. Then (C2') suffices for the *computational* problem: the reduction rebuilds $\text{Inst}(\tau_i(\vec{x}_0))$ for each i as in the proof of Theorem 1, queries the solver on all m of them to get $W_1, \dots, W_m$, and returns

$$\prod_{i=1}^m W_i^{c_i} \cdot [h(\vec{x}_0)]_\star = \left[\sum_{i=1}^m c_i f(\tau_i(\vec{x}_0)) + h(\vec{x}_0) \right]_\star = [f(\vec{x}_0)]_\star.$$

The step that makes this go through is that a $\mathbb{Z}_p$ -linear combination of *exponents* is realized by group operations on their encodings, so arbitrary linear post-processing of the solver's answers is available to the reduction for free. Encodings are transparent to it: the map $\vec{x} \mapsto [f(\vec{x})]_\star$ is a Reed–Muller codeword read through the encoding, and a linear local-decoding procedure for that code never notices the encoding is there. The whole Lipton [Liptong1] toolkit therefore applies to computational Uber problems verbatim.

Two costs come with $m > 1$. The reparametrizations τ_i must be correlated for the interpolation identity to hold — in Lipton's instantiation they are the translations along a single random line, uniform marginally but not jointly — so the reduction's success probability is controlled by a union bound rather than transferred exactly. And, as Section 10.3 records, the technique does not reach decisional problems at all.

10.2 Interpolation reaches a Type-3 target the affine technique cannot

Proposition 6 (interpolation in Type 3). *Let $p > 4$ and take the Type-3 instance of Proposition 5: $R = (1, X_1)$ in $\mathbb{G}_1$, $S = (1, X_2)$ in $\mathbb{G}_2$, $T = (1)$, and $f = X_1 X_2^2$ with challenge in $\mathbb{G}_T$. The computational $(R, S, T, f; \mathbb{G}_T)$ -Uber problem is random self-reducible by a non-adaptive four-query reduction $\mathcal{B}^{\mathcal{A}}$ satisfying, for every $\vec{x}_0 \in \mathbb{Z}_p^2$,*

$$\text{Succ}(\mathcal{B}; \vec{x}_0) \geq 1 - 4(1 - \text{Succ}_{\text{avg}}(\mathcal{A})).$$

Proof. Draw $\vec{b} = (b_1, b_2) \leftarrow_{\$} \mathbb{Z}_p^2$ and fix four distinct nonzero $t_1, \dots, t_4 \in \mathbb{Z}_p$, which exist as $p > 4$. Let $\tau_i: \vec{x} \mapsto \vec{x} + t_i \vec{b}$.

Rebuild. Each τ_i is a translation, so $X_1 \circ \tau_i = X_1 + t_i b_1 \in \mathcal{L}_1 = \langle 1, X_1 \rangle$ and $X_2 \circ \tau_i = X_2 + t_i b_2 \in \mathcal{L}_2 = \langle 1, X_2 \rangle$, which is (C1); the $\mathbb{G}_T$ tuple is the constant 1 and needs nothing. Explicitly $\text{Inst}(\vec{x}_0 + t_i \vec{b}) = ([1]_1, [x_1]_1 [1]_1^{t_i b_1}, [1]_2, [x_2]_2 [1]_2^{t_i b_2}, [1]_T)$, computable from $\text{Inst}(\vec{x}_0)$ with no pairing and no ψ .

Uniformity. For each fixed i , $t_i \neq 0$ and $\vec{b}$ uniform make $\vec{x}_0 + t_i \vec{b}$ uniform on $\mathbb{Z}_p^2$, which is (C3) for each τ_i separately. The four are not jointly independent: they lie on one line.

Interpolate. Put $g(t) := f(\vec{x}_0 + t\vec{b}) = (x_1 + t b_1)(x_2 + t b_2)^2$, a polynomial in t of degree at most 3, and let $c_i := \prod_{j \neq i} (-t_j)(t_i - t_j)^{-1}$ be the Lagrange coefficients at 0, so that $q(0) = \sum_i c_i q(t_i)$ for every q of degree at most 3. This is (C2') with $m = 4$ and $h = 0$. Query $W_i \leftarrow \mathcal{A}(\text{Inst}(\vec{x}_0 + t_i \vec{b}))$ and output $Y := \prod_{i=1}^4 W_i^{c_i}$. If every $W_i = [g(t_i)]_T$ then

$$Y = \left[\sum_{i=1}^4 c_i g(t_i) \right]_T = [g(0)]_T = [f(\vec{x}_0)]_T.$$

Each query is marginally a fresh average-case instance, so $\mathcal{A}$ errs on any fixed one with probability $1 - \text{Succ}_{\text{avg}}(\mathcal{A})$, and a union bound over the four gives the stated bound. $\square$

The reduction never computes $[x_2^2]_T$, the element Proposition 5 shows to be out of reach. It does not have to: it never rebuilds the covariance defect, because the solver returns that defect implicitly inside its own answers, and recovering $f(\vec{x}_0)$ from those answers is a $\mathbb{Z}_p$ -linear operation the encoding permits.

10.3 The decisional problem stays blocked

Remark 3. Interpolation does not transfer to the decisional problem of Proposition 5. There the reduction must map its own challenge Z to a challenge Z_i for each rebuilt instance, so it needs the encodings of the coefficients of

$$g(t) = \underbrace{x_1 x_2^2}_{t^0} + \underbrace{(b_1 x_2^2 + 2b_2 x_1 x_2)}_{t^1} t + \underbrace{(b_2^2 x_1 + 2b_1 b_2 x_2)}_{t^2} t^2 + \underbrace{b_1 b_2^2}_{t^3} t^3.$$

The t^2 and t^3 coefficients lie in $\mathcal{L}_T = \langle 1, X_1, X_2, X_1 X_2 \rangle$, but the t^1 coefficient carries $b_1 X_2^2$, and $X_2^2 \notin \mathcal{L}_T$ in Type 3. Setting $b_1 = 0$ removes it at the cost of freezing the first coordinate, destroying (C3) exactly as in the proof of Proposition 5. The asymmetry is structural rather than incidental: interpolation runs from the solver's answers to the target, which is linear and hence free in the exponent, whereas a decisional reduction must also run from the target to the solver's challenges, and that direction needs the intermediate coefficients themselves.

10.4 What each technique buys

The two reductions are incomparable, and the axes trade against each other. Theorem 1 admits every $\tau \in \text{AGL}_n(\mathbb{Z}_p)$, including the multiplicative scaling that Proposition 4 shows to be forced for co-CDH and that no reduction restricted to translations along a line could supply; interpolation admits any f of degree less than p whose line restriction rebuilds, with no covariance requirement at all. In exchange, Theorem 1 transfers advantage exactly, in both the search and the decisional problem, whereas Proposition 6 needs $\text{Succ}_{\text{avg}}(\mathcal{A}) > 3/4$ before its bound says anything and says nothing about decisional problems. The threshold is an artifact of naive interpolation rather than of the approach — Berlekamp–Welch decoding and its list-decoding successors push it down to a constant and then to $\text{poly}(1/\epsilon)$ [GS92, BTo6] — but no version of it reaches negligible advantage, and none reaches a decisional problem. Cryptographic assumptions are stated overwhelmingly in the decisional form and at negligible advantage, which is precisely the regime where the covariance route is the only one available; that is why Theorem 1, and not interpolation, is the tool used throughout this note.

Two further points of contact are worth recording. The translation reduction for CDH and DDH is the group-theoretic form of Blum–Luby–Rubinfeld self-correction [BLR93]: the shift $\vec{x} \mapsto \vec{x} + \vec{b}$ is their shift, with the group law supplying the linearity their argument needs. And the worst-case-to-average-case reductions of lattice

cryptography, from Ajtai [Ajtai96] to Regev [Regev09], share the motivation of this note — basing an average-case assumption on a worst-case one — but are reductions *between distinct problems* rather than self-reductions, and their mechanism, Gaussian sampling over a lattice, has nothing to do with the polynomial structure exploited here.

11 Discussion

Across the four settings the reduction of Theorem 1 is one and the same; only the admissibility conditions move, and they move only because the computable target space $\mathcal{L}_\star$ grows with the available products:

$$\text{none} \subsetneq R \cdot S \text{ (Type 3)} \subsetneq R \cdot S \cup S \cdot S \text{ (Type 2)} \subseteq R \cdot R \text{ (Type 1, single group)}.$$

A larger $\mathcal{L}_\star$ makes the covariance condition (C2) easier to meet, so the affine-self-reducible subclass is largest in Type 1 and smallest in Type 3, with the non-bilinear case the degenerate product-free floor. Proposition 5 witnesses the strict gap between Type 2 and Type 3 for the affine technique; by Proposition 6 the gap closes for that target once several correlated queries are allowed and only the computational problem is asked, so what the taxonomy of Lemma 1 constrains is the reach of tight and decisional reductions.

Self-reducibility is orthogonal to hardness. The master theorem of [BBG05, Boyen08] makes an assumption hard exactly when $f \notin \mathcal{L}_\star$, whereas Theorem 1 needs the defect $f \circ \tau - \lambda_\tau f$ to lie in $\mathcal{L}_\star$. These constrain different objects, so hard yet self-reducible assumptions abound: CDH and DDH (non-bilinear), BDDH (Type 1), $X_1 X_2^2$ (Type 2), and co-CDH (Type 3) are all of this kind. Quantitatively, Bauer, Fuchsbauer, and Loss [BFL20] show that in the algebraic group model every member of the Uber family is implied by the q -discrete-logarithm assumption, with q governed by the degrees of the defining polynomials, and Rotem [Rotem22] refines q to the maximal degree in which a single variable occurs and gives improved bilinear reductions.

12 Bibliography

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